GENERAL SPECIFICATION G4.26

Lamella Settlement System

CONTENTS

G4.26 LAMELLA SETTLEMENT SYSTEM

G4.26.1 Equipment Description

(a) Plate Pack Frames / Plate Support System

(i) Plate pack frames and plate support structures shall be constructed of Type 316L stainless steel. Structural members shall be suitable for long-term service with continuous immersion in the process water.

(ii) Plate pack frames shall be designed with anchorage baseplates at each end of the plate pack frame modules. The anchorage at one end of the frame shall be designed for fixed end conditions, and the other anchorage shall provide for movement that may result from contraction/expansion of the plate pack frame and/or the concrete structure housing the parallel plate settlers. Frames shall be designed to span end to end support.

(iii) Each frame/support shall be factory fabricated and shall be delivered to the Site in one piece. No field fabrication of frames/supports will be permitted. The Contractor shall also furnish all corrosion resistant accessories required for mounting of frames/supports on concrete walls.

(iv) The frames or plate supports shall be structurally designed to span between the concrete supporting walls and shall support all the weight of the plates, and water and sludge loads expected. Frame/support deflection shall be limited to 1/360th of the span. The frames/supports shall be designed to withstand hydrostatic uplift created when filling the settlement basin and shall support live loads. All loading and structural design shall be designed and endorsed by a P.E. of the structural discipline and submitted to the S.O. for approval.

(b) Inlet Feed Baffles

(i) Inlet feed baffles and flumes shall be constructed of Type 316L stainless steel suitable for long-term service with continuous immersion in the process water and exposure to the atmosphere.

(ii) The inlet shall be hydraulically designed to equally distribute the influent to all the plates. Properly sized openings in the side walls of the plates shall introduce the flocculated water near the bottom of the plates from both sides, and minimise floc destruction. Computational Fluid Dynamics (CFD) modelling shall be conducted to demonstrate that equal hydraulic distribution is achieved and the CFD modelling report shall be submitted to the S.O. for approval. Any modification work required based on modelling shall be subjected to S.O.’s approval.

(c) Parallel Plates

(i) The parallel plate assemblies shall be fabricated from Type 316L stainless steel of at least 0.7 mm thick, stiffened at top with structural bend and crimp.

(ii) The parallel plate assemblies shall be secured in the plate pack frames in accordance with the manufacturer’s standards.

(iii) The plates shall have sufficient wetted length-to-width ratio of greater than or equal to 2:1 to give good distribution of flow across the lower end of the plates.

(iv) Plates shall be easily replaceable individually.

(d) Outlet Weirs/Flumes

(i) Outlet weirs and flumes shall be straight-edged weirs constructed of Type 316L stainless steel.

(ii) 45° V-notch weir plate shall be factory pre-assembled onto the top of each outlet flume side. The weirs shall be designed to allow adjustment in the field to level all weirs to the same elevation. The tops of the V-notch weirs shall be flattened to remove sharp edges that pose a safety hazard.

(iii) The weirs/flumes shall provide a means of effluent flow balancing to ensure even flow distribution through the plates.

(e) Scraper

(i) The rotating scraper assembly shall comprise of two fabricated steel arms manufactured from Rectangular Hollow Section (R.H.S.) to which are attached steel scrapers positioned at an angle to the arms. The settled sludge shall be moved towards the centre sludge hopper or sump of each settlement tank. The equipment manufacturer shall ensure minimal maintenance works will be needed for the scrapper.

(ii) The scraper shall be suitable for installation in lamella settlement tanks having floor slope.

(iii) The scraper shall be centre pier supported, and of centre drive type.

(iv) Furnish complete scraper assembly including drive motor, gearing pedestal, mounting plate, bridge, walkway, platforms, handrail and other necessary parts, including anchor bolts.

(v) Direction of mechanism rotation shall be clockwise when viewed at from the top.

(vi) The two scraper arms shall be securely attached to and rotated by a central fabricated steel drive tube and drive plate. Adjustable steel tie rods shall be fitted between the drive tube and scraper arms to support the weight of the cantilevered arms and provide adjustment for levelling on installation.

(vii) A rigid steel coupling shall be securely attached to the vertical output shaft of the secondary gearbox providing the connection between the top flange of the drive tube and drive unit.

(viii) A fabricated steel bridge structure if specified, shall be supported above top water level in recesses in the outer walls of the settlement chamber, shall be provided to carry the drive unit and scraper arm assembly and be complete with anodised aluminium hand railing and galvanised open mesh flooring over half its length to provide access to the drive unit. Hand railing at all corners shall be curve-bend. 90 degree sharp-bend will not be acceptable. All supporting steelwork shall be secured to the concrete by stainless steel fixings. Resin anchors shall not be used. The steel bridge shall be located such that removal of lamella plates will not be obstructed.

(ix) The scraper shall be driven by a totally enclosed weather-proof electric motor of minimum IP55 rating and complete with anti-condensation heaters. The motor shall be horizontally flange mounted to a primary heavy duty co-axial helical gear reducer and attached to the final helical worm drive unit fastened to an adapter plate on the support bridge.

(x) The rotational speed of the scraper arms necessary to give the required peripheral speed of the scraper blades shall be provided by a final worm gear reducer. This unit shall be complete with a heavy duty extended bearing and shaft to provide the required stability for the scraper assembly.

(xi) The scraper blades shall be fitted to the underside of the scraper arm to give a scraping action over the tank floor from the outer wall to a point approximately past the top edge of the sludge sump.

(xii) Each scraper blade assembly shall have a flexible blade clamped to the rear of a support plate by a backing strip.

(xiii) The scraper blade shall be set at 60 degrees to the tank radius and shall ride the undulations of the tank floor as the scraper arms rotate.

(xiv) Scraper blade dimensions shall be not less than 150 mm wide x 20 mm thick, with 55 mm protruding below the bottom edge of the support plate.

(xv) Scraper blades shall be fixed onto the support plate by means of fixing holes and fasteners along the centre of the blade so that the blade may be inverted to double its working life, which shall not be less than 20, 000 hrs.

(xvi) The gear units shall be connected by a combined flexible coupling and torque limiter with automatic reset. An electrical limit switch operated by the torque limiter shall be positioned to disconnect the drive and protect the equipment in the event of a blockage. Both the coupling and limiter shall have a read out at Local Control Panel according to the relevant sections of the General Specification on Electrical and ICA disciplines related items.

(f) Scum Removal

(i) For Used Water (wastewater) applications a proven scum removal facility shall be provided and shall connect to the scum hopper if specified.

(g) Plate Cleaning

(i) A proven automatic washing and cleaning system shall be provided to periodically clean the lamella plates without affecting the normal operation of the plant.

G4.26.2 Materials of Construction

(a) Lamella Plates: Stainless steel ASTM 316L.

(b) Scraper arm and accessories (including clamps, fixings, dowels, pins, etc.): Stainless steel ASTM 316L.

(c) Scraper blade: High quality, wear resistant rubber, or polyurethane, specially compounded to resist wear and damage.

(d) Fasteners: Shall conform to ASTM A193 and ASTM A194, Type 316 B8MN, B8M2, or B8M3. Fasteners shall be threaded in accordance with ANSI B1.1 for screw threads, coarse-thread series. Stainless steel fasteners lubricant (anti-seizing) shall be applied to the threads prior to making up the connections

(e) Welding

(i) All welding shall be done in accordance with the recommendations of the American Welding Society;

(ii) All welding shall be done by a process suitable for the materials to be welded;

(iii) Welds shall be free of porosity, cracks, holes, and flux;

(iv) All welds shall be ground smooth and shall have a uniform appearance; and

(v) Field welds shall be subjected to S.O.'s approval and shall be passivated prior to equipment being placed in service.

G4.26.3 Accessories

(a) Equipment Identification Plates: Provide 2 mm thick, Type 316 stainless steel identification plate securely mounted in a readily visible location on each separate equipment component and control panel(s). Plate shall bear 10 mm high engraved block type black enamel filled equipment identification numbers and letters.

(b) Lifting Lugs: Provide for equipment and components weighing over 25 kg.

(c) Anchor Bolts: Type 316L stainless steel as minimum.

G4.26.4 Manufacturer's Qualifications

(a) The materials covered by these specifications are intended to be standard equipment of proven reliability and as manufactured by a reputable manufacturer having experience in the production of such equipment. The equipment furnished shall be designed, constructed, and installed in accordance with the best practices and methods and shall operate satisfactorily when installed as shown in the Drawings and operated per Manufacturer's recommendations.

(b) The manufacturer shall have manufactured at least twenty comparable units which have been used for similar service each of which has been operating satisfactory for a minimum of four years.

(c) The manufacturer shall submit a past projects reference installation list with contact information for the equipment quoted verifying to the S.O. that the specified manufacturer's qualifications are met.

G4.26.5 Source Quality Control

(a) Each equipment shall be fully inspected and functionally tested at the factory for performance, vibration and proper operation.

(b) Factory Inspections: Inspect for required construction, electrical connection, and intended function.

(c) Factory Test Report: Provide test reports certified correct by the manufacturer.

G4.26.6 Factory Finishing

(a) Prepare, prime, and finish coat with the following systems, as specified in the General Specification-Painting and Protective Coatings.

(i) System No. 19 or 20 as appropriate according to the specific material of construction for all submerged metal surfaces;

(ii) System No. 18 for exposed metal surfaces; and

(iii) No painting required for stainless steel surfaces.

G4.26.7 Preparation for Shipment

(a) Insofar as is practical, the plate pack modules shall be shipped assembled, ready for installation at the Site. Parts and assemblies that are, of necessity, shipped unassembled shall be trial assembled at the factory and marked in a manner to facilitate final assembly in the field. The equipment and materials shall be packaged in a manner that will protect the equipment from damage during shipment.

(b) Contractor shall have a manufacturer’s representative at the Site during receipt of equipment and materials. This representative together with a representative of Contractor and S.O. shall inspect all equipment and materials upon arrival at the Site. Damaged or otherwise unacceptable equipment and materials shall be removed from the Site and replaced with new equipment and materials. Accepted equipment and materials shall be turned over to Contractor for storage in accordance with the manufacturer’s instructions until installation is carried out.

(c) Equipment and materials provided under this section shall be delivered and inspected. No equipment shall be shipped until the factory test results have been accepted by the S.O..

G4.26.8 Installation

(a) A detailed description of the various items of work and precaution in handling the plate settling units shall be provided by the manufacturer to the Contractor prior to delivery and installation.

(b) Provide a specially fabricated sling for use with individual plate settling units before installation commences. Sling shall be the property of the Board at the end of the installation for plant maintenance usage.

(c) Installation work shall conform to manufacturer’s recommended procedures, instructions, and shop drawings.

(d) The Contractor shall notify the S.O. of all activities of inspection, transportation, receipt, unloading, storage, handling and protection of equipment in advance of the actual works in weekly and other progress look-ahead meetings.

(e) Provide supervision, labour, tools, construction equipment, incidental materials, and necessary services required for equipment installation.

(f) Installation of the lamella settlement tank system shall not begin prior to satisfactory completion of supporting structures. Support structure columns, beams, walls or slabs shall not be used to move the equipment into position.

G4.26.9 Field Quality Control

(a) Functional Test: Prior to plant start up, all equipment described in this Specification shall be inspected for proper alignment, connection and satisfactory performance by means of a functional test as performed by Contractor.

(b) Performance Test

(i) Conduct on each unit;

(ii) Perform under actual or approved simulated operating conditions;

(iii) Test for a continuous 3 hours period without malfunction;

(iv) Perform with the S.O.'s present for test witnessing;

(v) Test Log: Upon completion of test, record information listed on sample test log; and

(vi) Adjust, realign, or modify units and retest if necessary.

(c) Electrical equipment shall be tested in accordance with General Specification G5.51 Electrical Testing.

G4.26.10 Manufacturers' Services

(a) Provide onsite services of the entire duration of the manufacturer's representative for the following activities:

(i) Installation assistance, supervision, and inspection;

(ii) All pre-commissioning checks;

(iii) Functional and performance testing;

(iv) Completion of Manufacturer's Certificate of Proper Installation Compliance and

(v) Commissioning assistance.

G4.26.11 Manufacturer's Certificates

(a) The following certificates shall be provided:

(i) Certification of Proper Installation;

(ii) Certification of Materials Compliance;

(iii) Certification of Functional Test; and

(iv) Certification of Performance Test

END OF SECTION